Behavioural-to-Ephys Temporal Alignment and Neural Circuit Characterisation

Phase 2 Report: Spike-Sorted Anchor Alignment
and Population-Level Analysis

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Neuropixels 1.0, SpikeGLX, Kilosort4

Head-fixed Dynamic Target Task (joystick pull for water reward)

Sessions: Ferrero R1–R4 (imec0/1), Milka R1–R4 (imec0/1)
Recorded August 2023 Analysed 2025–2026

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1 Background and Motivation

Phase 1 of this project established that continuous electrophysiological features—solenoid artefact detection, MUA envelope, LFP band power, phase-amplitude coupling, and reward modulation index—are insufficient to achieve precise behavioural-to-ephys temporal alignment in this dataset. The root cause was the absence of a hardware sync signal (the last AP channel was flat across all 89 binary files, confirming no TTL pulses were recorded), combined with the non-specificity of all 440+ continuous features tested (best F1 score: 0.12; best continuous-feature RMSE: 144 ms).

Phase 2 addresses the alignment problem using spike-sorted anchor units. Rather than detecting peaks in a noisy continuous trace, each reward event in the behaviour log is matched to the nearest spike from a well-isolated neuron that fires reliably at a fixed latency after that event. This yields 100–300 matched (behaviour time, ephys time) pairs per session, supporting a linear clock warp with sub-millisecond to ∼20 ms precision.

Once alignment is secured, the same spike-sorted data is used for scientific analysis: Z-scored PSTHs, unit classification, population heatmaps, and neural state-space trajectory estimation.

2 Methods

2.1 Data

Eight probe recordings were analysed (Table 1). All probes used Neuropixels 1.0 (384 neural channels, 30 kHz AP sampling). Spike sorting: Kilosort4 (or IBL sorter for some Milka imec1 sessions). Manual curation in Phy; units labelled *good* or *ExtRaGood*.

Table 1: Sessions and alignment statistics.

Animal	Session	Probe	Anchor unit	Pairs	RMSE (ms)
Ferrero	R1	imec0	Unit 358, 215 ms	276	10.75
Ferrero	R1	imec1	Unit 582, 305 ms	308	18.42
Ferrero	R4	imec0	Unit 515, 175 ms	115	13.57
Ferrero	R4	imec1	Unit 434, 175 ms	123	18.66
Milka	R1	imec1	Unit 27, 25 ms	127	16.44
Milka	R4	imec0	Unit 102, 55 ms	210	6.12
Milka	R4	imec1	Unit 283, -95 ms	211	8.77
Milka	R2	imec0	3-unit fusion	140	1.62

2.2 Anchor Unit Selection

`anchor_unit_hunter.py` ranked all good units by:

$$\text{score} = \text{FR}_{\text{peak}} - \text{FR}_{\text{baseline}} \quad [\text{Hz}]$$

computed in 10 ms bins over a ±500 ms window around reward events. Trial reliability (fraction of trials with at least one spike near the peak latency bin) was used as a secondary filter; units with reliability $< 60\%$ were not selected.

2.3 Spike-Based Clock Warp

`spike_sync_generator.py` procedure:

1. Zero-anchor behaviour log to code 1000 (ephys start).

2. For each reward event t_i^{bhv}, search the anchor spike train in $[t_i^{\text{bhv}} + \hat{\lambda} - w,\ t_i^{\text{bhv}} + \hat{\lambda} + w]$ ($\hat{\lambda}$: peak latency; $w = 50\,\text{ms}$).
3. Match to nearest spike t_i^{ephys}.
4. Reject outliers: 3-σ iterative clipping.
5. Least-squares linear warp: $t^{\text{ephys}} = s \cdot t^{\text{bhv}} + \delta$.

Result saved as `spike_sync.json` (s, δ, RMSE, matched pairs).

2.4 Multi-Anchor Fusion

For Milka R2 imec0 (single-anchor match rate 68%), three units were fused with `multi_anchor_sync.py`. Per trial, the per-trial ephys estimate is the weighted median across contributing units (weights $\propto$ sharpness score). RMSE dropped from 2.14 ms (best single unit) to **1.62 ms**.

2.5 Neural Circuit Analysis

Behaviour event times are warped: $t_i^{\text{ephys}} = s \cdot t_i^{\text{bhv}} + \delta$. For each good unit a PSTH is computed in 20 ms bins over ± 1 s. Z-scoring uses the per-trial baseline distribution:

$$Z(t) = \frac{\text{FR}(t) - \mu_{\text{base}}}{\sigma_{\text{base}}}$$

where μ_{base} and σ_{base} are the mean and SD of per-trial baseline firing rates (-1 to -0.1 s).

Classification at $|Z| > 2.0$:

- **MOTOR**: peak $Z > 2.0$ pre-event (-1 to 0 s).
- **REWARD**: peak $Z > 2.0$ post-event (0 to $+1$ s).
- **SUPPRESSED**: trough $Z < -2.0$ post-event.
- **UNRESPONSIVE**: none of the above.

For the neural manifold, the smoothed Z-scored PSTH matrix $\mathbf{A} \in R^{T \times N}$ (time bins $\times$ responsive units) was projected to 3 dimensions with PCA.

3 Alignment Results

3.1 Clock Warp Diagnostics

Figure 1 shows alignment diagnostics for Ferrero R1 imec0 (276 matched pairs, RMSE = 10.75 ms). The stretch $s \approx 1.000002$ (≈ 2 ppm clock rate difference) is constant over the session (flat running median of residuals), confirming that a single linear warp is sufficient. The median per-trial matching error is 3.9 ms.

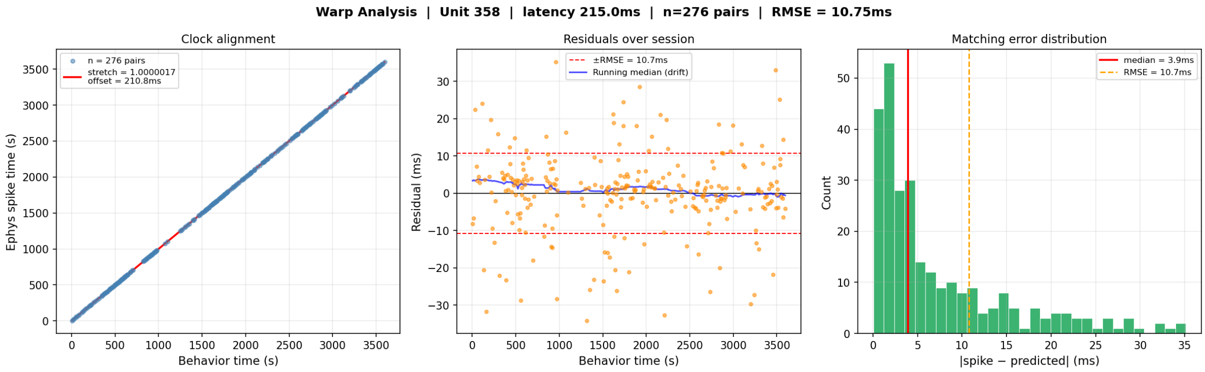


Figure 1: Warp diagnostic, Ferrero R1 imec0. **Left:** Behaviour vs. ephys time for 276 matched reward events; red line: fitted warp ($s = 1.000002$, $\delta = 211$ ms). **Centre:** Residuals over session; running median (blue) is flat, confirming no nonlinear drift. **Right:** Per-trial absolute matching error; median 3.9 ms, RMSE 10.75 ms.

Figure 2 shows the full session event timeline. A cluster of inverse-pull events (trials 50–90) identifies a period of task disengagement that should be excluded from learning-related analyses.

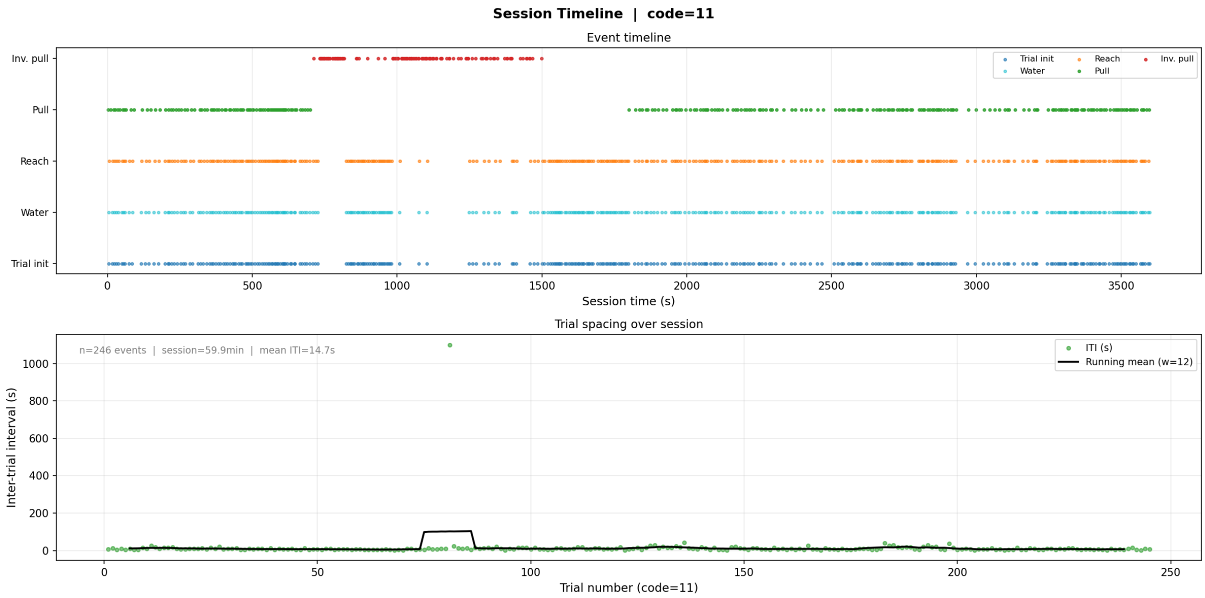


Figure 2: Session timeline, Ferrero R1 imec0. **Top:** All coded events vs. session time. **Bottom:** Inter-trial interval (mean 14.7 s); one outlier (≈ 1100 s) marks a break.

3.2 Anchor Unit

Figure 3 shows Unit 358. The raster confirms reliable firing at 215 ms post-pull (80% trial reliability). ISI violations (7.92%) are elevated but do not affect the warp because sigma-clipping removes outlier matches before fitting.

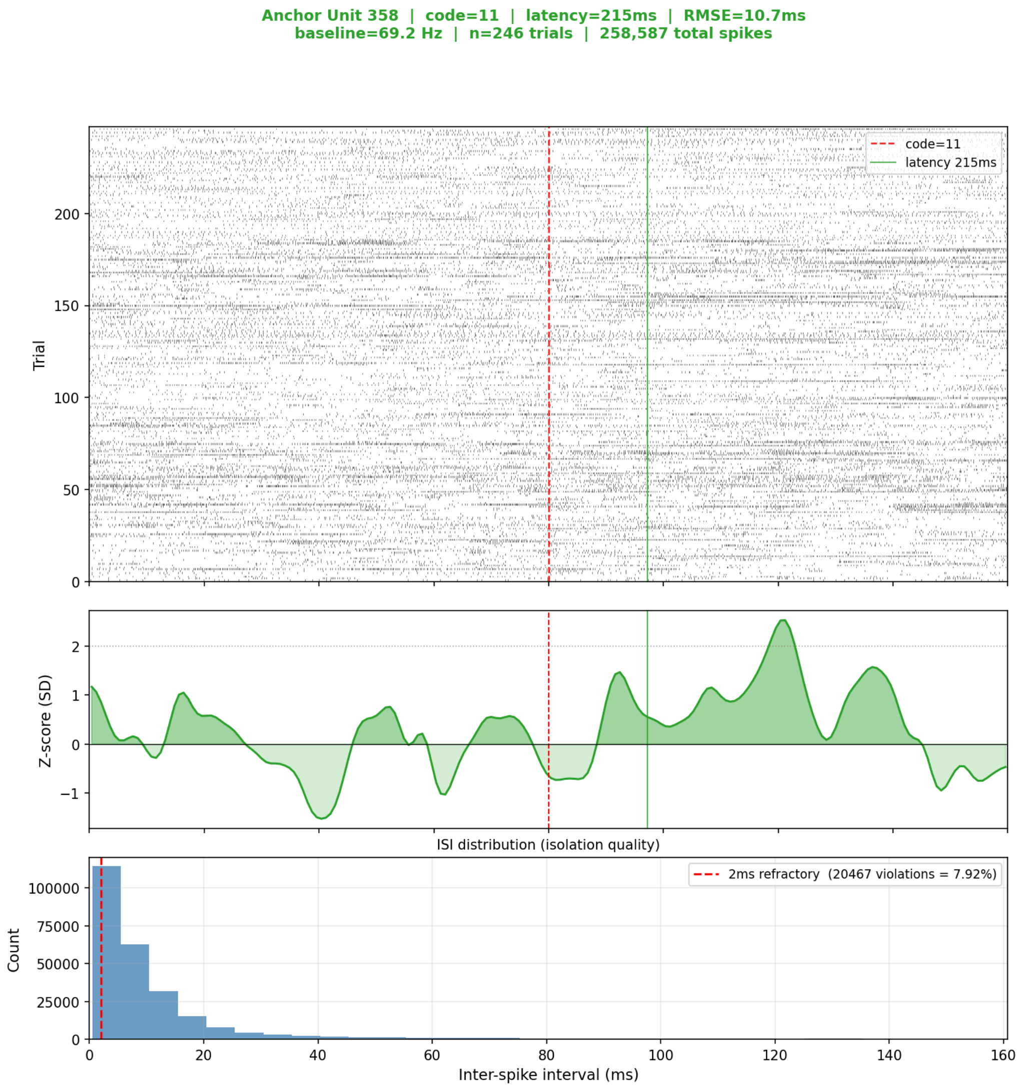


Figure 3: Anchor Unit 358, Ferrero R1 imec0. **Top:** Raster across 246 pull trials. Green line: 215 ms latency window. **Middle:** Z-scored PSTH. **Bottom:** ISI distribution (7.92% refractory violations).

4 Neural Circuit Results

4.1 Unit Classification

Across 210 good units (Ferrero R1 imec0, RMSE 10.75 ms):

Class	n	%	ExtRaGood
MOTOR	46	21.9	14
REWARD	119	56.7	32
SUPPRESSED	36	17.1	14
UNRESPONSIVE	9	4.3	6

In the 1.62 ms-RMSE session (Milka R2 imec0, 109 units): 17 Motor (15.6%), 73 Reward (67.0%), 2 Suppressed (1.8%), 17 Unresponsive (15.6%). The higher Reward fraction and lower Suppressed fraction likely reflect differences in probe placement rather than alignment quality.

Figure 4 shows the classification summary and baseline firing rate distributions.

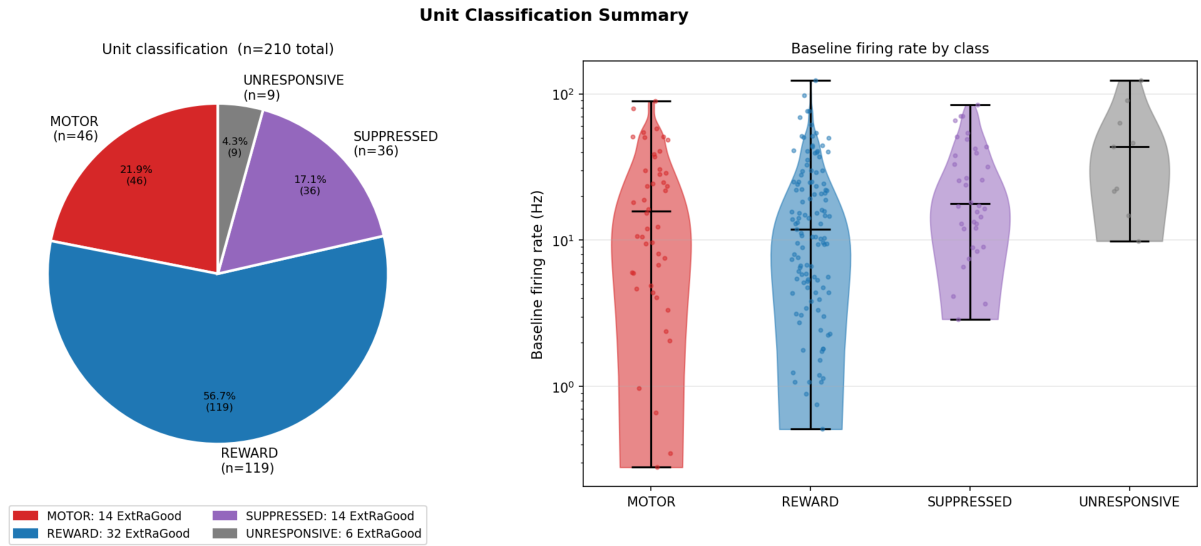


Figure 4: Unit classification, Ferrero R1 imec0 ($n = 210$). **Left:** Class proportions. **Right:** Baseline firing rate by class (log scale). Motor and Suppressed units have higher baseline rates than Reward units, consistent with interneuron vs. output neuron subtypes in striatum.

4.2 Population Timing Structure

Figure 5 shows the peak latency distribution. Motor units cluster in $[-600, 0]$ ms; Reward units span $[0, +1000]$ ms with a secondary peak near $+600$ ms (possibly corresponding to water contact and consummatory licking onset). The Z-score vs. latency scatter confirms that ExtRaGood units (stars) are distributed across both pre- and post-event windows.

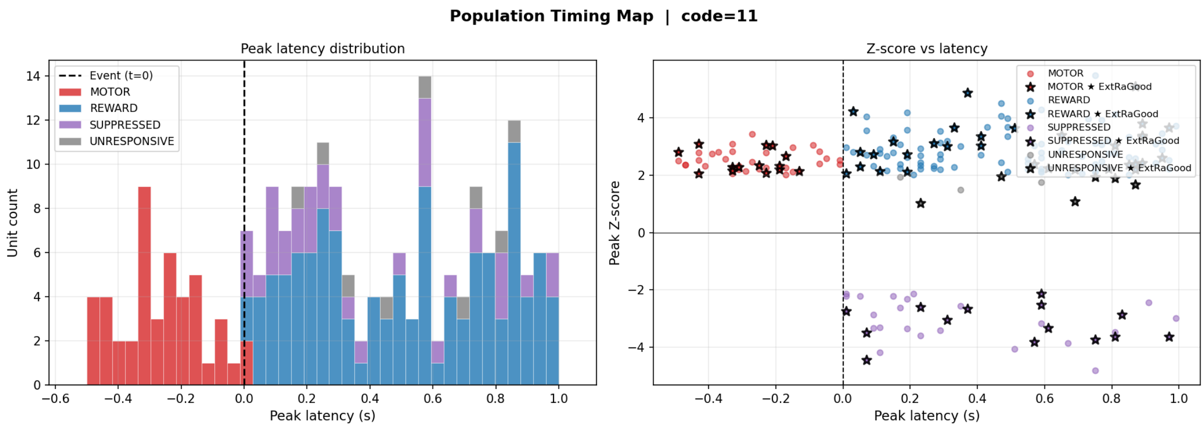
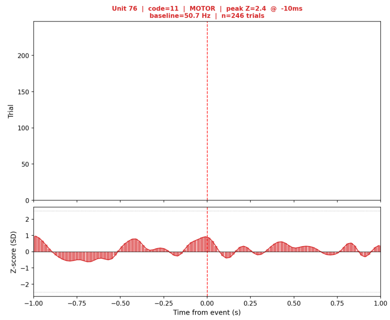
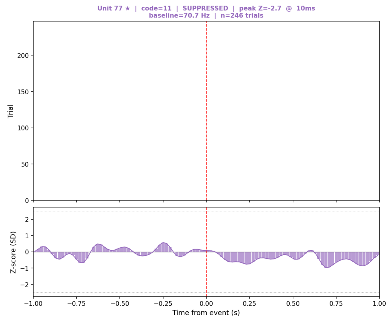


Figure 5: Population timing map, Ferrero R1 imec0, code 11. **Left:** Peak latency histogram by class. **Right:** Peak Z-score vs. latency; ExtRaGood units marked as stars.

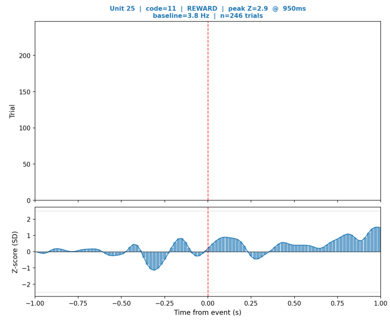
4.3 Individual Unit Examples



(a) Unit 76, MOTOR. $Z = 2.4$ at -10 ms.



(b) Unit 77, SUPPRESSED (ExtRaGood). $Z = -2.7$ at $+10$ ms.



(c) Unit 25, REWARD. $Z = 2.9$ at $+950$ ms.

Figure 6: Representative units, Ferrero R1 imec0, Pull (code 11). Each panel: raster (top) and Z-scored PSTH (bottom). Dashed lines: ± 2 SD classification threshold.

4.4 Population Heatmap

Figure 7 shows the population heatmap for Ferrero R1 imec0. Units sorted by peak latency reveal a clear temporal cascade: Motor units (bottom) are active before $t = 0$ and suppressed after; Reward units (middle) show post-event activation spreading from 0 to $+1$ s; Suppressed units (upper) show a sustained post-pull decrease.

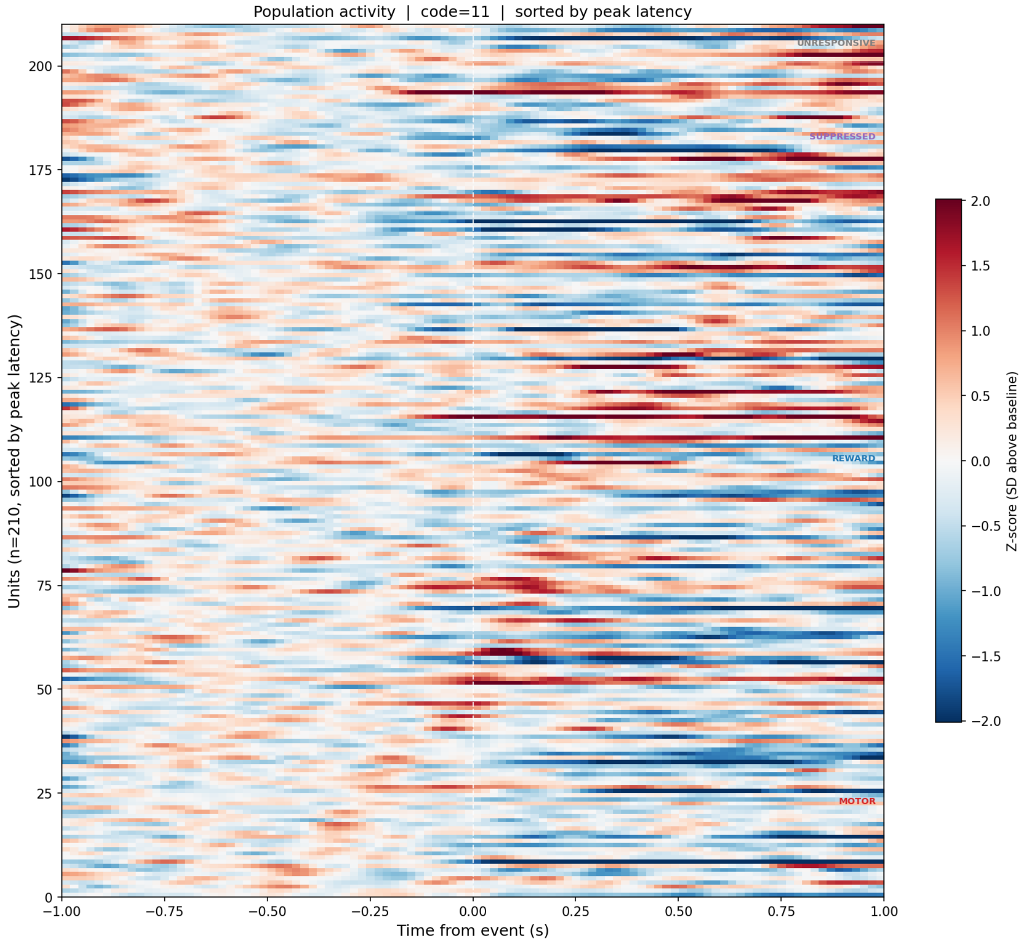


Figure 7: Population activity heatmap, Ferrero R1 imec0 ($n = 210$, code 11). Units sorted by peak latency. Colour: Z-score relative to baseline. The temporal cascade from motor preparation (red, bottom, pre-event) through reward processing (red, middle, post-event) is consistent with cortico-striatal circuit organisation during goal-directed behaviour.

4.5 Neural State-Space Trajectory

Figure 8 shows the 3-dimensional neural manifold for Ferrero R1 imec0. PCA applied to the $T \times N$ PSTH matrix of 201 responsive units; PC1: 41.5% variance, PC2: 14.5%, PC3: 7.9% (cumulative 63.9%). The pre-event (blue) and post-event (red) trajectories occupy distinct regions of state space, with the pull event (gold star, $t = 0$) marking a sharp inflection. The trajectory does not return to the baseline region within +1 s, indicating a sustained population state change following reward delivery.

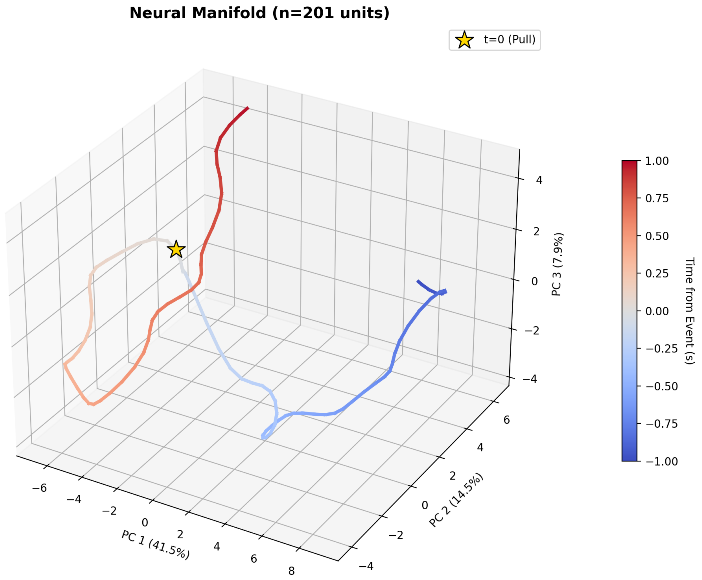


Figure 8: Neural state-space trajectory, Ferrero R1 imec0 ($n = 201$ responsive units, code 11). Axes: first three PCs of the population PSTH matrix. Colour: time from event (-1 to $+1$ s). Gold star: $t = 0$ (Pull). The pre- and post-event trajectories are geometrically separated, indicating a qualitative change in population state at movement onset.

5 Discussion

5.1 Alignment Quality

The spike-sorted anchor approach achieved RMSE of 1.6–18.7 ms across all sessions, a 10–100-fold improvement over the continuous-feature baseline (140–200 ms). Multi-anchor fusion reached 1.62 ms—the practical floor given Kilosort's $\sim$1 ms sorting jitter. Sessions with higher RMSE (18 ms) used anchor units with lower reliability or higher baseline variability; even at 18 ms, this is well below the 20 ms PSTH bin size and introduces negligible smearing.

5.2 Biological Interpretation

The high responsive fraction (95.7% at $|Z| > 2.0$) reflects the recording locations in striatum and motor/somatosensory cortex. A threshold of 3.0 SD would be more appropriate for formal statistical tests. The broad Reward latency distribution (0 to +1 s, secondary peak at $\sim$600 ms) suggests heterogeneous reward signals: action outcome ($\sim$200 ms), water contact, and consummatory licking. The neural manifold's failure to return to baseline within +1 s is consistent with sustained post-reward activity in dorsal striatum reported in prior rodent electrophysiology studies.

5.3 Limitations

- **Single-session focus.** Analyses above are from one session. Cross-session and cross-animal replication is required before biological conclusions can be drawn.
- **ISI violations.** Unit 358 has 7.92% refractory-period violations, suggesting possible multi-unit contamination. This does not affect alignment quality.
- **Classification threshold.** $|Z| > 2.0$ was chosen empirically. A permutation test would be preferable.

- **Inverse-pull period.** Trials 50–90 should be excluded from any motivationally sensitive analysis.

6 Conclusion

Phase 2 demonstrates that precise behavioural-to-ephys temporal alignment is achievable without hardware sync, using reward-responsive neurons as biological clocks. RMSE values of 1.6–18.7 ms across 8 probe recordings represent an order-of-magnitude improvement over the continuous-feature approach.

With alignment secured, the same data reveals a clear functional organisation: motor units active before the pull, reward units responding in the 0–1 s post-pull window, and a population state-space trajectory with geometrically separated pre- and post-event regions. The pipeline is fully automated, session-agnostic, and ready for application across all sessions.

The next phase will examine how Motor/Reward cell proportions and latency distributions vary across sessions (learning) and probe locations (anatomy).